

Computer Simulation

Lesson 5: Graphics

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Special thanks to: Dr. Pham Thi Huong

Graphics

- Introduction
- Plot Function
- Multiple Graphs
- Graphics Annotation
- Customizing Plots
- Graphics Properties

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Introduction

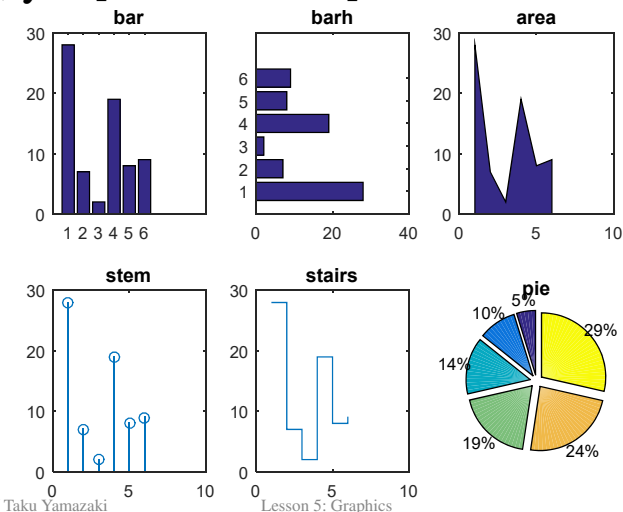
- MATLAB is not only a useful calculation tool but also well known as a powerful graphical application
- It is easy to create and customize
- MATLAB supports many kind of plot types
 - Line plots: linear (plot, plot3), log_log, semi_log, error bar
 - Pie, bar, histograms
 - Discrete data plots: stem, stair, scatter
 - Area plots, polar plots
 - Animation

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Plot Types Example

$x = 1:6; y = [28 \ 7 \ 2 \ 19 \ 8 \ 9]$



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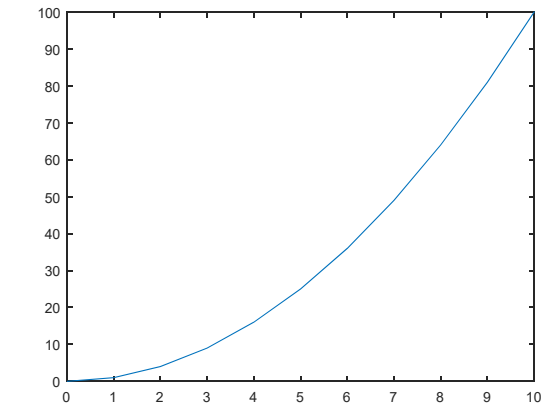
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The Plot Function

- **help** graph2d/graph3d
- `plot(x,y)`: linear plot vector y versus vector x

The Plot Function: Example

- `x=0:10;y=x.^2;plot(x,y)`
- Try to create another graph: `plot(x, x.^3)`
→???

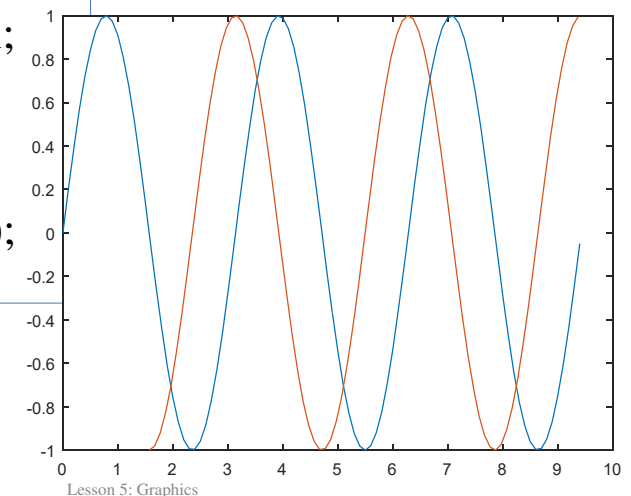


Multiple Graphs

- `plot(x1,y1,x2,y2)`: create multiple graphs in the same figure window
- Function **hold**: create multiple graphs in the same figure window
 - Clear the current graph when calling to the other plot function (default)
 - Add a new graph to the current graph (hold on)
- Function **figure**: create multiple graphics in different figure windows.
 - The order of the figure windows will be the order when they are created (default) or specified by typing: **figure(order_number)**
- Function **subplot**: create multiple graphs in tiled positions of the Figure window

Multiple Graphs: plot

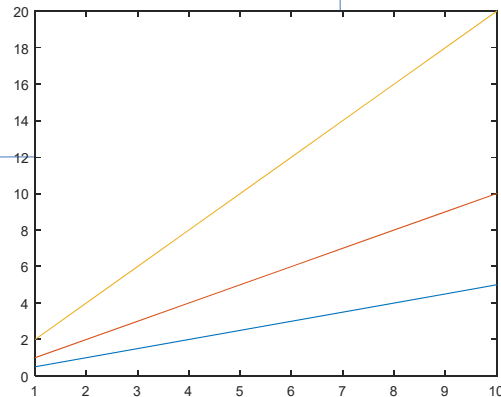
```
x1 = 0:0.1:3*pi;  
x2 = pi/2:0.1:3*pi;  
y1 = sin(2.*x1);  
y2 = cos(2.*x2);  
plot (x1,y1,x2,y2);
```



Multiple Graphs: hold

```
x=1:10; y1=1/2.*x; y2=x; y3=2.*x; y4=3.*x;
plot(x,y1); hold on
plot(x,y2)
plot(x,y3); hold off
plot(x,y4)
```

% after the command,
when a new graph is
created, MATLAB will
clear all current graphs

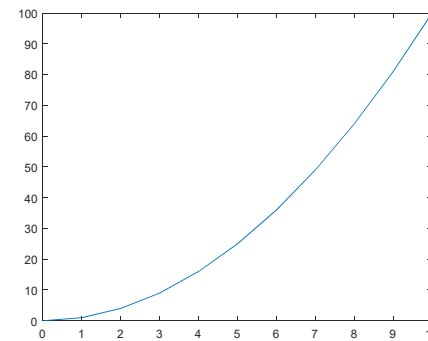


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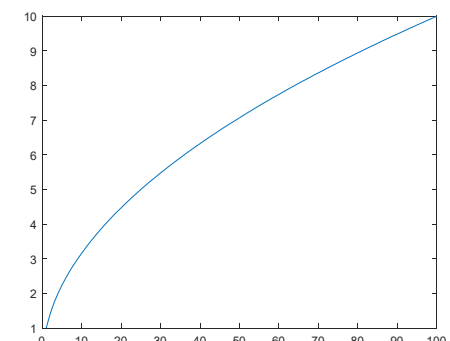
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Multiple Graphs: figure

- Example: `x=1:10; y=x.^2; plot(x,y)`
`x=1:100; y=sqrt(x); figure; plot(x,y)`
 Or `x=1:100; y=sqrt(x); figure(2); plot(x,y)`
- Graphs will be deleted when the figure windows are closed



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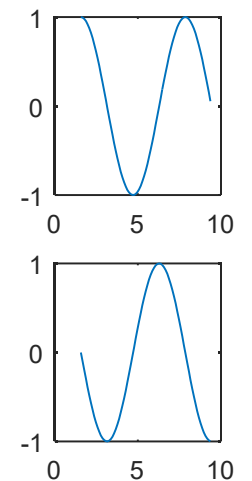
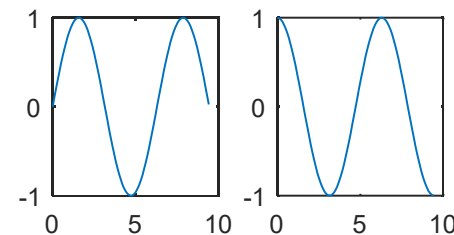
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Multiple Graphs: subplot

- MATLAB numbers its subplots by row
- Example: `subplot(3,4,p)`

```
subplot(3,4,1)
x1=0:0.1:3*pi; plot(x1,sin(x1))
subplot(3,4,2)
plot(x1,cos(x1))
subplot(3,4,8)
x2=pi/2:0.1:3*pi; plot(x2,sin(x2))
subplot(3,4,12)
plot(x2,cos(x2))
```

3,4,1	3,4,2	3,4,3	3,4,4
3,4,5			
			3,4,12



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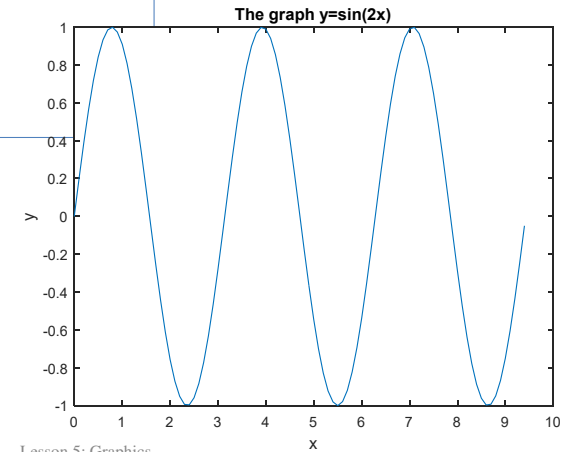
Graph Annotation

- title: graph title
- xlabel: horizontal x-axis label
- ylabel: vertical y-axis label
- text: text annotation
- gtext: place text with mouse
- legend: display legend

Graph Annotation: title and labels

```
x = 0:0.1:3*pi; y = sin(2.*x);  
plot (x,y);  
title('The graph y=sin(2x)');  
xlabel('x'); ylabel('y');
```

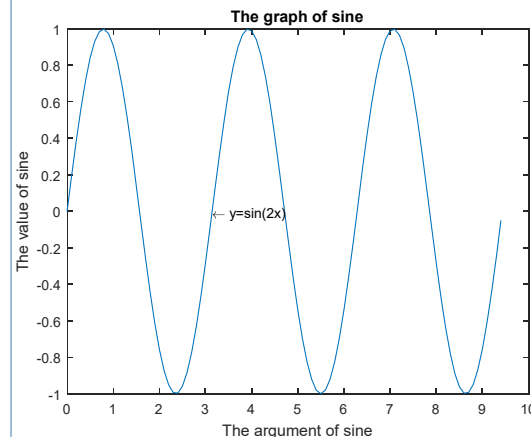
help xlabel,
ylabel, title



Graph Annotation: text annotation

Text annotation adds the text in the quotes to a specific location on the current axes

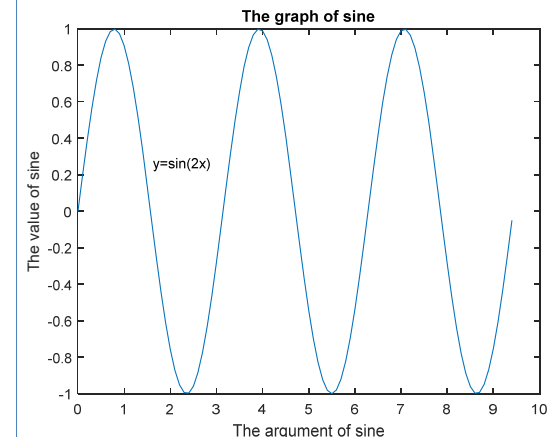
```
x = 0:0.1:3*pi;  
y = sin(2.*x);  
plot (x,y);  
title('The graph of sine');  
xlabel('The argument of sine');  
ylabel('The value of sine');  
text(pi,0,'← y=sin(2x)');
```



Graph Annotation: gtext

gtext: mouse placement of text in 2-D view

```
x = 0:0.1:3*pi;  
y = sin(2.*x);  
plot (x,y);  
title('The graph of sine');  
xlabel('The argument of sine');  
ylabel('The value of sine');  
gtext(' y=sin(2x)');
```

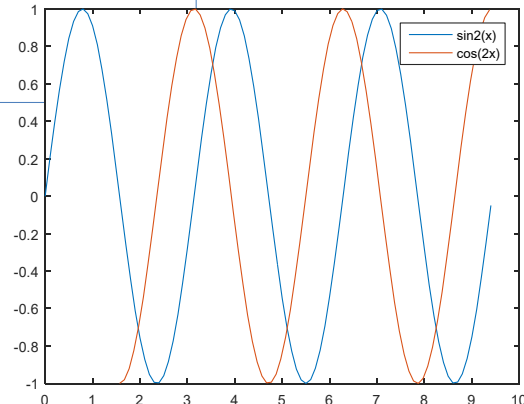


Graph Annotation: legend

Add legend to graph

```
x1 = 0:0.1:3*pi; x2 = pi/2:0.1:3*pi;  
y1 = sin(2.*x1); y2 = cos(2.*x2);  
plot(x1,y1,x2,y2);  
legend('sin2(x)','cos(2x)')
```

legend
legend off



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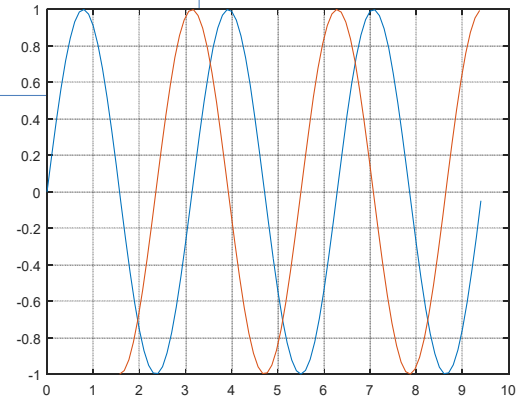
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Customizing Plots: grid

Display or hide the axes grid lines

```
x1 = 0:0.1:3*pi; x2 = pi/2:0.1:3*pi;  
y1 = sin(2.*x1); y2 = cos(2.*x2);  
plot(x1,y1,x2,y2);  
grid on
```

Grid on
Grid off
Grid minor



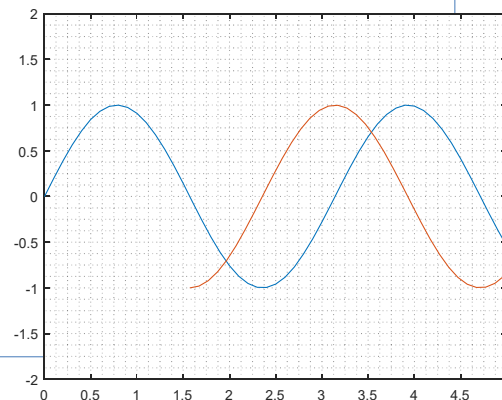
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Customizing Plots: axis

- axis: control axis scaling and appearance
 - axes: create axes in arbitrary positions
 - Example: Set axis limits for the current axes
- ```
axis([xmin xmax ymin ymax])
```

```
x1 = 0:0.1:3*pi;
x2 = pi/2:0.1:3*pi;
y1 = sin(2.*x1);
y2 = cos(2.*x2);
plot(x1,y1,x2,y2);
axis([0 5 -2 2])
```



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## Graphics Properties (Line style, color, width, marker properties)

- Line properties
  - Line style
  - Line width
  - Line color
- Marker properties
  - Marker style
  - Marker size
  - Marker color

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# Graphics Properties: Colors

| String | Color   | [RGB] |
|--------|---------|-------|
| b      | blue    | 001   |
| c      | cyan    | 011   |
| g      | green   | 010   |
| b      | black   | 000   |
| m      | magenta | 101   |
| r      | red     | 100   |
| y      | yellow  | 110   |
| w      | white   | 111   |

Color properties also can be specified by matrices: [RGB]

# Graphics Properties: Line Style

| Symbol | Description     |
|--------|-----------------|
| --     | dashed          |
| -.     | dash dot        |
| :      | dotted          |
| -      | Solid (default) |
| none   | No line         |

# Graphics Properties: Markers

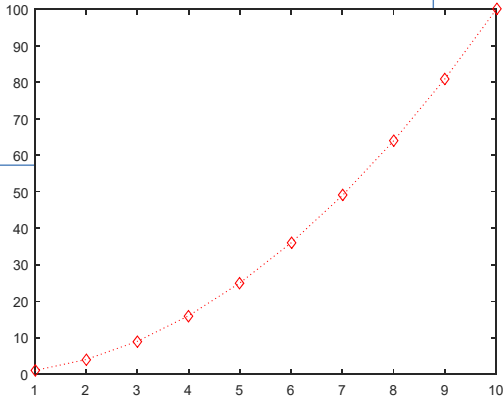
| Symbol | Description    |
|--------|----------------|
| o      | Circle         |
| d      | Diamond        |
| h      | Hexagram       |
| p      | Pentagram      |
| +      | Plus           |
| .      | Point          |
| s      | Square         |
| *      | Star           |
| v      | Down triangle  |
| <      | Left triangle  |
| >      | Right triangle |
| ^      | Up triangle    |
| x      | X-mark         |

Default: no marker

# Example

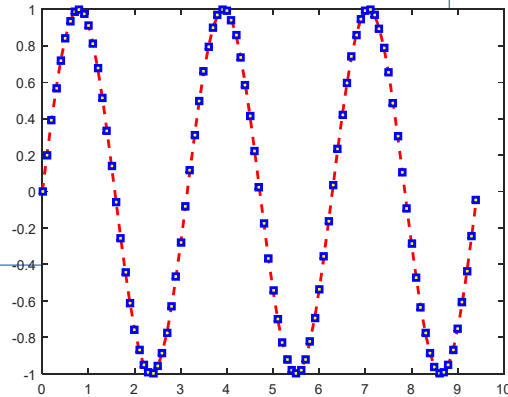
- `plot(x,y, 'S')`: plot vector Y versus vector X with line properties are described in string S
- Example:  
`plot(1:10,(1:10).^2,'rd:')`

Line properties 'rd:'  
Line color: r → red  
Maker: d → diamond  
Line style: : → dotted



## Additional Graphic Properties

```
x = 0:0.1:3*pi; y = sin(2.*x);
plot(x,y,'rs--',...
 'LineWidth',2,...
 'MarkerEdgeColor','b',...
 'MarkerFaceColor','y',...
 'MarkerSize',5)
```



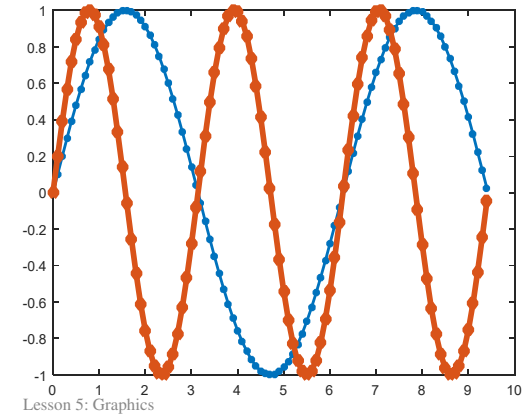
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## The dot notation (.)

- To set properties for graphs → useful to modify graphs after creation %Lesson5plotdot.m

```
x = 0:0.1:3*pi; y1=sin(x); y2=sin(2.*x);
p=plot(x,y1,x,y2);
p(1).LineWidth=2;
p(1).MarkerSize=5;
p(1).Marker='*';
p(2).LineWidth=4;
p(2).MarkerSize=5;
p(2).Marker='d';
```



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## Practice 5.1

- Write a script to display information about average temperatures of a country or city in 1 year
  - Ask user to press one key 1, 2, or 3 to display information
    - 1: highest temperature
    - 2: lowest temperature
    - 3: all temperatures

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Thank you very much for your attendance!

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# Related Plot Functions

| Function | Brief description                                                                                    |
|----------|------------------------------------------------------------------------------------------------------|
| clf      | Clear the Figure Window                                                                              |
| figure   | Create a new Figure Window.<br>figure(n) → refer to the n-th figure window                           |
| hold     | Create multiple graphs in the same Figure Window that allow superimposing (hold on, hold off)        |
| legend   | Displays a legend box on a graph                                                                     |
| grid     | Display grid line on a graph<br>grid on: turn on the grid lines<br>grid off: turn off the grid lines |